

S Meena Padnekar

RESEARCH SCHOLAR

Visualization and Perception Lab, Indian Institute of Technology, Madras, India

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Profile

PhD researcher at IIT Madras working on semantic scene understanding under sparse and adverse conditions. My research focuses on learning semantic and geometric structure with limited supervision, particularly through multimodal perception, sparse-view 3D Gaussian Splatting, and reconstruction under challenging sensing conditions. I work under the guidance of Prof. Sukhendu Das and Prof. Kaushik Mitra.

Research Interests

Computer Vision, 3D/4D Vision, Scene Understanding, Unsupervised Semantic Segmentation, Sparse-View Reconstruction, Neural Rendering, Multimodal Learning, Optimization, and Deep Learning.

Education

Doctor of Philosophy

CGPA: 8.53

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

January 2022 - Present

- Thesis: *Semantic Scene Understanding under Sparse and Adverse Conditions*
- Coursework: Computer Vision, Machine Learning, Deep Learning, Digital Image Processing, Linear Algebra and Probability, Optimization Techniques

Master of Technology in Computer and Information Science

CGPA: 9.41

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

July 2019 - May 2021

- Dissertation Title: *Stance Prediction for Fake News Detection*
- Coursework: Machine Learning Algorithms, Neural Network, Deep Learning, Digital Image Processing, Statistics and Probability, Data Mining, Probabilistic Graphical Model

Bachelor of Technology in Computer Science and Engineering

CGPA: 8.06

MODEL ENGINEERING COLLEGE

July 2014 - June. 2018

- Btech Project: Hand Gesture Recognition Using Deep Learning

Research and Teaching Experience

Research Scholar

Chennai, India

DEPARTMENT OF COMPUTER SCIENCE, INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Jan. 2022 - Present

- Conduct research on semantic scene understanding under limited supervision, with emphasis on adverse sensing conditions, multimodal learning, sparse-view 3D reconstruction, and 4D vision.
- Design and train PyTorch-based research pipelines for segmentation, multimodal representation learning, neural rendering, and Gaussian Splatting.
- Work on method development, experimental design, benchmarking, ablation analysis, and technical writing for journal and conference publications.
- Contributed to an industry-linked vision project on vehicle analysis using image-based estimation methods.

Teaching Assistant

Chennai, India

VISUALIZATION AND PERCEPTION LAB, INDIAN INSTITUTE OF TECHNOLOGY MADRAS

Jan. 2022 - Present

- Served as TA for Computer Vision, Pattern Recognition and Machine Learning, Operating Systems, Introduction to Programming, Video Signal Processing, and Optimization for Computer Vision.
- Supported course delivery through demos, doubt-clearing sessions, assignment design, evaluation, and logistics.
- Recipient of the Star TA Award twice for teaching contributions in PRML and Video Signal Processing.

Work Experience

Wisilica India Pvt.Ltd

Kochi, India

SOFTWARE ENGINEER

Jun. 2018 - July. 2019

- Worked on Wisilica Lighting and Tracking project.
- Responsible for designing and implementing the existing functionalities to the latest technologies.
- Roles and Responsibilities:
 - Development of back-end web services and RESTful APIs for the product.
 - Involved in designing and developing required relational databases using MySQL and PostgreSQL.
 - Setup, manage and monitor servers to ensure continuity and production.
 - Developed analytic skills and attention to detail by analyzing the system log.
 - Responsible for web development using AngularJS, HTML, CSS.
 - Track bug using Redmine and Github.

- I was part of a team to study, design, and develop efficient backup software.

Technical Skills

- **Programming:** Python, PyTorch, NumPy, OpenCV, Data handling pipelines
- **Deep Learning & Computer Vision:** segmentation, depth estimation, self/contrastive learning, knowledge distillation; strong familiarity with modern CV architectures (CNNs/ViTs)
- **3D Vision / Neural Rendering:** 3D Gaussian Splatting (3DGS), sparse-view reconstruction, multi-view consistency, scene understanding
- **Training & Scaling:** mixed precision, Distributed Data Parallel (DDP) basics, checkpointing, experiment tracking (W&B/MLflow), reproducible configurations (Hydra-style workflows)
- **Systems / Infra:** Linux, SLURM/HPC workflows, GPU debugging and performance profiling
- **Tools:** Git
- **Previously Used:** C/C++, JavaScript; HTML/CSS, JQuery, AngularJS, REST APIs, WebSockets; MySQL, PostgreSQL, Redis, SQLite, Hadoop

Publications

- **Semantic-Guided 3D Gaussian Splatting for Sparse View Reconstruction and Segmentation**
S. Meena Padnekar, Kaushik Mitra, Sukhendu Das
Accepted at ICPR 2026.
- **Distilling Auxiliary RGB-T Features for Unsupervised Semantic Segmentation**
S. Meena Padnekar, Kaushik Mitra, Sukhendu Das
Image and Vision Computing, Volume 170, 2026, 105977. <https://doi.org/10.1016/j.imavis.2026.105977>
- **BiLSTM-Autoencoder Architecture for Stance Prediction**
S. M. Padnekar, G. S. Kumar, P. Deepak
2020 International Conference on Data Science and Engineering (ICDSE), Kochi, India, 2020. DOI 10.1109/ICDSE50459.2020.9310133

Awards and Academic Support

- Recipient of an €880 **travel grant** for participation in the 28th International Conference on Pattern Recognition (ICPR) 2026, France.
- Recipient of the **Star TA Award** twice for teaching contributions in PRML and Video Signal Processing.
- **Half-Time Research Assistant Scholarship**, IIT Madras.
- **GATE Scholarship**, 2019-2021.

Selected Projects

Deblurring and 4D Reconstruction using a single blurry monocular video and event stream

RESEARCH PROJECT 4D COMPUTER VISION

Jan 2026 - Present

- Investigating reconstruction under severe motion blur using event information and monocular visual input.
- Focus includes dynamic scene modeling, blur-aware reconstruction, and event-guided spatiotemporal reasoning.

Semantic-Guided 3D Gaussian Splatting for Sparse View Reconstruction and Segmentation

RESEARCH PROJECT ON COMPUTER VISION

Jan 2025

- Developed **SegSplat**, a few-shot 3D Gaussian Splatting framework for joint sparse-view reconstruction and semantic segmentation.
- Introduced a distilled prototype contrastive loss, geometry-conditioned diffusion-based pseudo-view augmentation, and semantics-aware Gaussian density control.
- Achieved state-of-the-art sparse-view novel view synthesis performance, including **22.84 PSNR on Tanks and Temples with 3 views**, and improved novel-view segmentation by up to **+9.22 mIoU**.

Unsupervised Segmentation of Images taken under Adverse Lighting Conditions using RGB-Thermal Imagery

RESEARCH PROJECT ON COMPUTER VISION

May 2024

- Developed **DARTS**, a multimodal unsupervised semantic segmentation framework for low-light, night-time, and glare-affected scenes.
- Designed a dual-modal ViT fusion block with self-attention and cross-attention, along with intra-modal and inter-modal feature-correlation objectives.
- Improved STEGO by **+2.36, +2.22, +4.87, and +0.67 mIoU on KP, MFNet, PST900, and SemanticRT** respectively.

Vehicle Wheel Alignment

VEHICLE WHEEL ALIGNMENT FROM STEREO IMAGES

March 2023 - March 2025

- Worked on estimating wheel orientation from a pair of stereo images using classical computer vision methods and appropriate image capture design.

Conferences and Workshops Attended

- Presented research work at the 2020 International Conference on Data Science and Engineering.
- Attended Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP), 2024.
- Attended Summer School on AI, IIIT Hyderabad, 2022.
- Attended ACM Winter School on Natural Language Processing, 2021.
- Attended CSA Undergraduate Summer School, Indian Institute of Science, Bangalore, 2016.

Strengths

- **Coding Skills** Proficient in developing, implementing, and debugging algorithms and research-oriented software solutions.
- **End-to-End Project Execution** Capable of driving projects through all stages, from concept to completion.
- **Teamwork and Collaboration** Strong ability to work effectively with teams in research and technical settings.
- **Research Manuscript Writing** Experienced in academic writing, paper development, and technical communication.
- **Leadership and Management** Proven ability to lead initiatives, manage responsibilities, and support team progress.
- **Teaching and Mentoring** Committed to knowledge sharing, student guidance, and collaborative learning.

References

Dr. Sukhendu Das, Professor, Department of Computer Science & Engineering, Indian Institute of Technology Madras, Email: sdas@cse.iitm.ac.in

Dr. Kaushik Mitra, Associate Professor, Department of Electrical Engineering, Indian Institute of Technology Madras, Email: kmitra@ee.iitm.ac.in

Dr. G Santhosh Kumar, Professor, Department of Computer Science, Cochin University of Science and Technology, Email: san@cusat.ac.in